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The Imaginary Lines

Let's Learn About

- Latitudes
- Longitudes
- Grid
- Greenwich Mean Time
- Indian Standard Time
- Maps

Warm Up

Look at the globe carefully. Notice the horizontal and vertical lines running on it. What are these lines called? Discuss in class.



The Earth's surface is made up of huge landmasses known as continents, water bodies known as oceans and seas, and a large number of small islands.

Geographers have created the globe for the purpose of locating places on the Earth. To do so, some imaginary lines are drawn horizontally and vertically on the globe. The vertical lines are called **longitudes** and the horizontal lines are called **latitudes**.

Learn More

The earliest known globe was made by a scholar named Crates around 140 BC.

Latitudes

(**Latitudes** are imaginary circles running from east to west on the globe. One such imaginary line of latitude is the **equator**. It divides the Earth into two equal halves known as **hemispheres**.) The northern part of the Earth above the equator is called the **Northern Hemisphere** and the southern part of the Earth below the equator is called the **Southern Hemisphere**. Latitudes are also called **parallels of latitude** as they run parallel to each other and to the equator.

Latitudes are measured with the unit **degree (°)**. The equator is the 0° latitude. Poles are 90° North and 90° South of the equator. The parallels in the Northern Hemisphere are labelled 10°N, 20°N, and so on. On the other hand, the parallels in the Southern Hemisphere are labelled 10°S, 20°S, etc.

There are 181 latitudes in all:

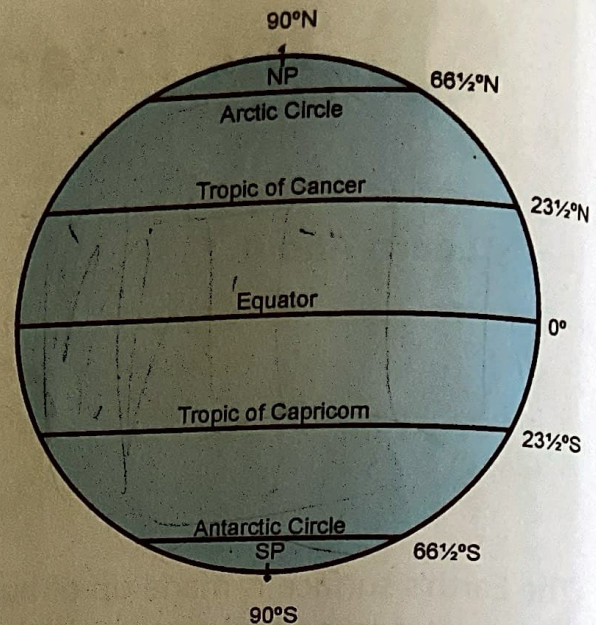
- Equator at 0°
- 90 latitudes from 1° to 90° in the Northern Hemisphere
- 90 latitudes from 1° to 90° in the Southern Hemisphere

Important Parallels of Latitudes

The most important and the longest parallel of latitude is the equator. All the other latitudes are numbered from it.

If we move northward from the equator, we will find the **Tropic of Cancer** at $23\frac{1}{2}^{\circ}\text{N}$, the **Arctic Circle** at $66\frac{1}{2}^{\circ}\text{N}$ and the **North Pole** at 90°N .

If we move southward from the equator, we will find the **Tropic of Capricorn** at $23\frac{1}{2}^{\circ}\text{S}$, the **Antarctic Circle** at $66\frac{1}{2}^{\circ}\text{S}$, and the **South Pole** at 90°S .



Characteristics of Latitudes

- The lines of latitudes are parallel to each other and they never meet.
- All latitudes are actually circles drawn in each hemisphere. They become smaller as they move towards the poles.
- All latitudes, other than equator, are called 'small circles' and they cannot divide the Earth into two equal halves.
- The North Pole at 90° in the Northern Hemisphere and the South Pole at 90° in the Southern Hemisphere are just two points on the tilted axis of the Earth.

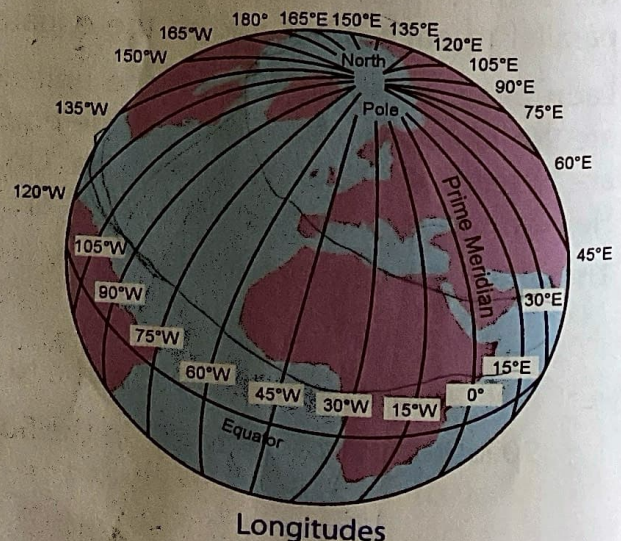
Learn More

Latitude of a place can be determined using an instrument called Sextant. Navigators carry this instrument to find their location over the sea.

Longitudes

(Longitudes are imaginary vertical lines drawn on a globe running from the North Pole to the South Pole. These are also known as **meridians** which means midday in Latin.)

The central longitude that divides the Earth into the Eastern Hemisphere and the Western Hemisphere is known as the **Prime Meridian**. It is the 0° meridian. This meridian runs through the observatory at Greenwich in London in the United Kingdom, and is also known as the **Greenwich Meridian**.)



All the other meridians are numbered from the Prime Meridian. There are 180 meridians to its east. They are marked as 1°E–179°E. The 180 meridians to the west of the Prime Meridian are marked as 1°W–179°W. The meridian of 180°E and 180°W is the same line. It passes through the islands of the Pacific Ocean. It is called the **International Date Line**. A new date begins when we cross this meridian.

There are 360 meridians in all.

- The Prime Meridian at 0°
- The International Date Line at 180°
- 179 meridians from 1°E to 179°E in the Eastern Hemisphere
- 179 meridians from 1°W to 179°W in the Western Hemisphere



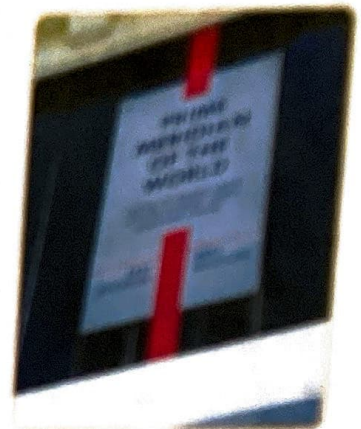
The International Date Line

Do and Learn

Look at the diagram of the International Date Line. Find out why it is zig-zag and not a straight line.

Characteristics of Meridians or Longitudes

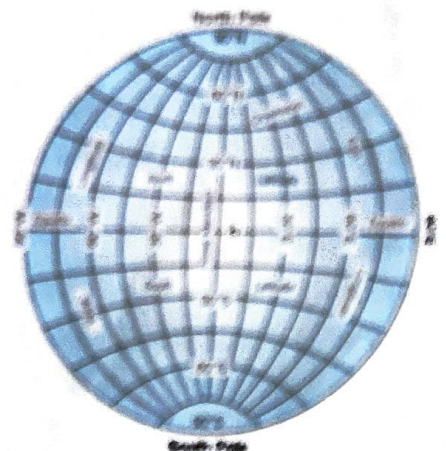
- ✓ All the meridians are of the same length and they meet at the poles.
- ✓ Meridians opposite to each other may form great circles and can divide the Earth into two equal halves.
- Due to the shape of the Earth, the distance between two longitudes is greater at the centre near the equator, and it keeps on decreasing as we move towards the poles.
- ✓ The longitudes to the east of the Prime Meridian are denoted with the symbol 'E' meaning east and the longitudes to the west of the Prime Meridian are marked with the symbol 'W' meaning west.



The Prime Meridian passing through the Royal Observatory at Greenwich

Grid

The latitudes and the longitudes intersect each other and form a network of lines called **grid**. It helps us to locate places accurately on the globe and maps. For this, we need to know the latitude and the longitude of the place. The point at which they cross each other will be the location of that place.



Grid on the globe

Greenwich Mean Time

The countries to the east of the Prime Meridian face the sun first. Gradually, as the Earth moves, the countries in the Western Hemisphere face the sun.

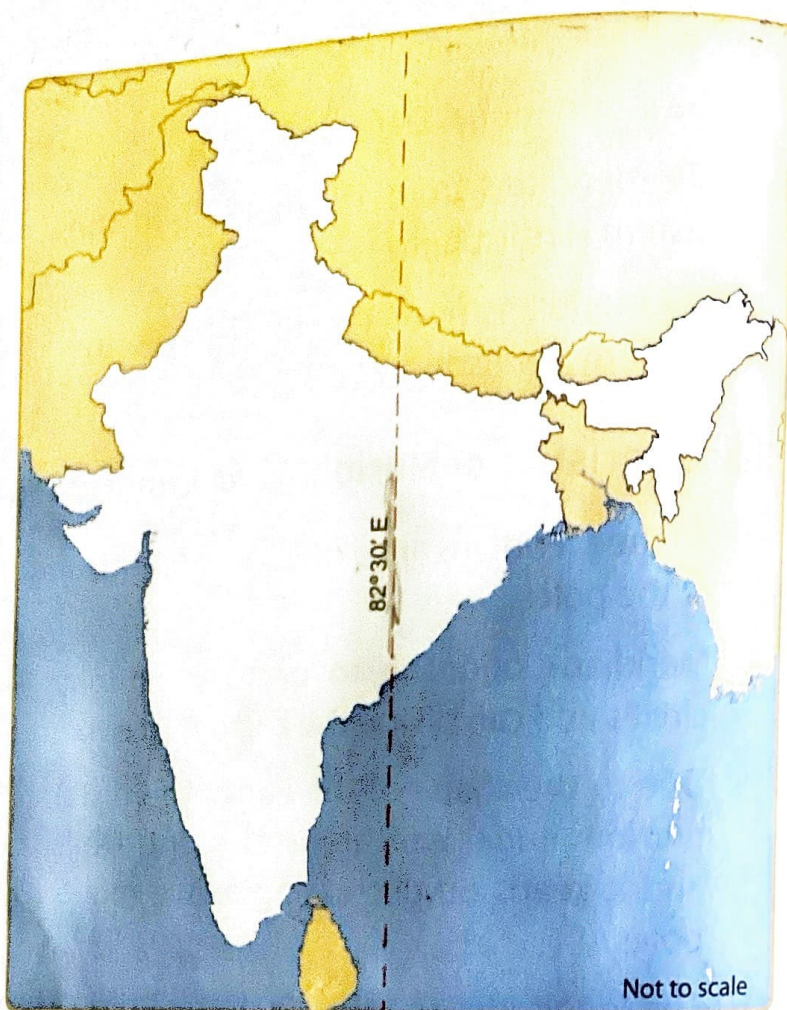
The time at Greenwich is observed as the mean time and hence the name **Greenwich Mean Time (GMT)**. Local time of a particular country can be calculated based on the GMT.

Learn More

Japan is known as 'the land of the rising sun'.

Indian Standard Time

A country may have many longitudes passing through it. Hence, different places may have different time within the country. Thus, every country chooses a longitude that passes through its centre and the time at that longitude is taken as the standard time of that country. In India, the longitude of $82\frac{1}{2}^{\circ}\text{E}$ is taken for calculating the standard time. This standard time is known as the **Indian Standard Time (IST)**. The Indian Standard Time is $5\frac{1}{2}$ hours ahead of the Greenwich Mean Time.



The Standard Meridian of India

Maps

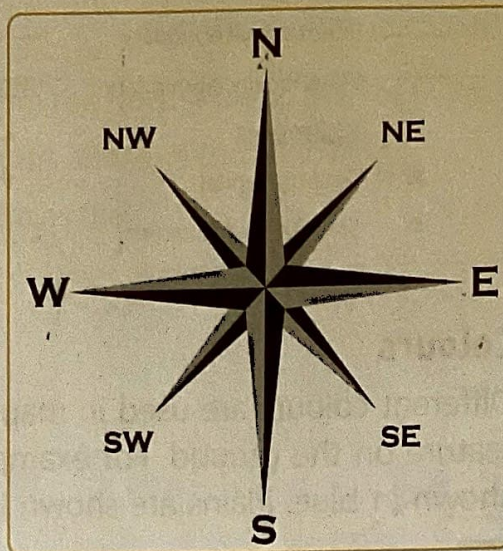
(A map is a visual representation of the Earth on a flat surface.) It can be considered as one of the oldest forms of communication. The art of making or drawing maps is called **cartography**. The oldest known map is a **Babylonian clay tablet**. (Maps can provide a lot of information about physical features, political divisions, weather, climate, agriculture, industries, minerals, trade route and many other things.) A book of maps is called an **atlas**.

Map Reading

We should know how to read a map in order to make full use of it. There is a certain kind of language associated with maps. There are different **symbols**, **colours** and **signs** to show different things in a map. Let's learn about some of them.

Directions

The first thing we should understand when we look at a map is the direction. In every map, the north direction is shown with the help of an arrow pointing upwards. When we place a map on a table, the top of the map shows the north direction. The bottom of the map is the south. The right side of the map is the east and the left side is the west. When we know the four main directions, we can easily locate other sub-directions also. For example, north-east lies between the north and the east; north-west lies between the north and the west; south-west lies between the south and the west; and south-east lies between the south and the east.



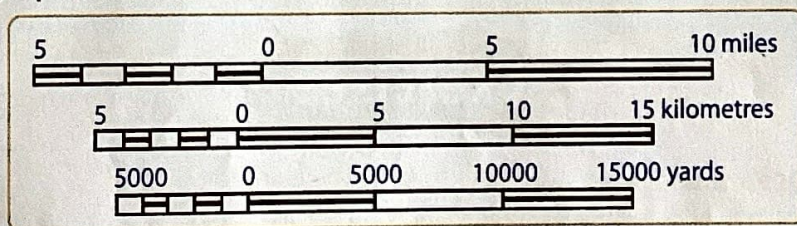
Directions

Scale

It is not possible to show the actual size of the things that are shown in maps. That is why scales are used. The scale is the ratio between two points on the ground and their corresponding distance on a map.

For example, suppose the distance between city A and city B is 500 km by road. It is not possible to draw that much distance in a map. Instead the distance of 500 km on the ground will be represented by a smaller unit (like 5 cm) in the map. So the scale would be 5 cm : 500 km.

Scale can be represented in three ways. They are representative fraction, written statement and graphic scale.



Scale

Class Discussion

Discuss the difference between a map with a scale and a map without a scale. Which one would be more helpful?

Symbols

Symbols represent some features on the map. They can be graphical or pictorial. There are symbols to show things like mountains, rivers, lakes, roads, bridges, airports, seaports and railway tracks.

Boundaries	Physical Features	Communications
International Boundary	River	National Highway with Number
State/Province Boundary	Lake/Reservoir	Other Road
Settlements	Peak (height in metres)	Railway
Country Capital	Sea depth (in metres)	Airport
State/UT Headquarters	Coral Reef	Seaport

Symbols in a map

Colours

Different colours are used in maps. Many of them have a relationship to the object or feature on the ground. For example, water bodies such as oceans, seas and rivers are shown in blue. Plains are shown in light green and deserts are shown in yellow.

	Blue	-	oceans, seas and rivers		Yellow	-	deserts
	Dark Green	-	forests		Light Green	-	plains
	Brown	-	mountains and hills		Red	-	plateaus

Let's Recall

- Latitudes and longitudes are the imaginary lines drawn on the globe.
- The Prime Meridian runs through Greenwich in London and the International Date Line passes through the islands of the Pacific Ocean.
- The Greenwich Mean Time is the time followed to find the time of different places at different longitudes.
- The Grid is the network of lines formed by the intersection of latitudes and longitudes.

Exercises

A Tick the correct answer.

- What are the horizontal lines on the globe called?
☐ Longitudes ☐ Latitudes ☐ Meridians
- What is the art of making or drawing maps called?
☐ Calligraphy ☐ Cartography ☐ Sketching
- Which of the following is the latitude that divides the Earth exactly into two halves?
☐ Arctic Circle ☐ Antarctic Circle ☐ Equator
- Which colour represents the water bodies on a map?
☐ Green ☐ Blue ☐ Yellow

B Fill in the blanks.

1. Latitudes are measured with the unit _____.
2. _____ are imaginary lines running from the North Pole to the South Pole.
3. The Indian Standard Time is _____ hours ahead of the GMT.
4. The network of latitudes and longitudes is known as the _____.

C Write (T) for true and (F) for false sentences.

1. All the meridians are of different lengths.
2. The lines of latitude are parallel to each other.
3. There are 180 latitudes in all.
4. There are 360 meridians in all.

D Answer the following questions.

1. What are latitudes?
2. What are longitudes?
3. What is a grid? How is it useful to us?
4. Define the term 'map'. What are the things that help us to read a map?

E Give reasons.

1. Latitudes are also called parallels of latitude.
2. The Prime Meridian is also called the Greenwich Mean Time.
3. Imaginary lines are drawn on the surface of the globe.

Activities

Tech Corner

(Creativity, Computer skills)

Using the Internet, find out the difference in the time of any five foreign countries in comparison to India. Make a note of your findings in MS Word/LibreOffice Writer document.

Creative Corner

(Arts Integration, In-depth learning, Creativity)

1. Draw a circle. Make the Prime Meridian dividing it into two equal halves. Colour and label the Eastern Hemisphere and the Western Hemisphere.
2. Draw another circle and make the equator dividing it into two halves. Now colour the Northern Hemisphere and the Southern Hemisphere.